Eric Balch

650-447-9086 | erichbalch@gmail.com | linkedin.com/in/eric-balch | ericbalch.github.io

Education

University of California, Los Angeles Sept. 2024 – June 2028
Bachelor of Science in Mechanical Engineering *GPA: 3.95/4.0*
- Extracurriculars: Bruin Formula SAE Racing, Bionics Lab, Tau Beta Pi

Experience

Mechanical Engineering Intern June 2026 – Sept 2026
ANELLO Photonics Inc. *Santa Clara, CA*
- Electromechanical design, development, assembly, & testing of Inertial Navigation System calibration fixtures
- Developed and released 16 custom machined and sheet-metal fixture elements through detailed design, structural analysis, and engineering drawings
- Selection & integration of high-precision motors and encoders for repeatable actuation
- Developed 2D part & assembly drawings w/ precise GD&T callouts for manufacturing and internal documentation
- Increased unit calibration throughput by 250% while maintaining mG level performance accuracy
- Designed breakout cable tester (KiCad) to validate harness function & streamline manufacturing quality control

Drivetrain Lead & Responsible Engineer Oct. 2024 – Present
Bruin Formula SAE Racing Drivetrain Team *Los Angeles, CA*
- Directing drivetrain architecture, leading team of 20 through full design cycle
- Designed (SolidWorks) front/rear sprockets and motor shaft w/ Design-for-Manufacture (DFM) emphasis
- Reduced mass by >10% & reduced rotational inertia by 33% through iterative nonlinear FEA testing (ANSYS) to optimize slotting
- Developed GD&T drawings & communicate specs to local sponsors for outsourcing component manufacturing
- Created and executed (F360 CAM) CNC toolpaths for manufacturing of motor shaft and motor mounts, reducing manufacturing costs by >25%
- Reduced in-house manufacturing time for drivetrain mounts by >20% by transitioning from soft jaws to tabbing
- Designed 80/20 assembly to stress test motor-to-sprocket chain (Up to 4500 N)

Undergraduate Researcher Nov. 2025 – Present
UCLA Bionics Laboratory *Los Angeles, CA*
- Design (SolidWorks) endoscope with independently controlled camera and surgical accessories to improve polyp resection and ease of use for surgeons
- Development of the housing and operational mechanism for the accessory attachment point
- Rapid prototyping of designs using 3D printed components & off-the-shelf tubing & control wires

Biomechanical Engineering Intern June 2023 – Sept. 2023
Stanford University *Stanford, CA*
- Developed computational fluid dynamics (CFD) simulation models of diseased coronary arteries
- Ran CFD simulations evaluating grafts with complex BCs to consider the effects of multiple stenoses
- Modeled pipeline for cardiovascular anatomic models of patient-specific coronary bypass procedures to analyze changing hemodynamics of different attachment locations

Lifeguard and Swim Instructor June 2021 – Sept. 2025
City of Mountain View *Mountain View, CA*
- Head Guard leadership role supervising a staff of up to 5 lifeguards, CPR & water rescue certified

Relevant Projects

Team Lead & Lead Designer, Engineering Lab Course Sept. 2022 – June 2024
- Designed and built winning wireless car parking lot sensors & display for CTE Capstone project
- Designed and built winning robots for two Vex Robotics timed competitive challenges
- Designed and built hydraulic arm to perform timed competitive precision challenges
- Coordinated team member contributions and presented product design results to class

Technical Skills

Software: SolidWorks, Fusion 360, ANSYS Mechanical, MS Office, Gemini, MATLAB, Python, Java, KiCad
Tools: Lathe & Mill (CNC & Manual), Drill Press, Band Saw, 3D Printer, Laser Cutter, Soldering